



Nothing Happens at Twelve: A Regression-Discontinuity Study of Express-Lane Item Limits, Service Time, and Basket Bunching Across 22 Supermarkets

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Abstract. The posted item limit on a supermarket express lane is a threshold rule of unusually clean form: a single integer, displayed at the point of sorting, applied by the shopper to herself. We treat it as such. Using lane-level point-of-sale records for 38,412 checkout transactions across 22 stores of one national grocery chain, we estimate a sharp regression discontinuity at the twelve-item limit, with basket item count as the running variable. The limit sorts effectively — the probability of using an express lane falls by 54.8 percentage points at the cutoff — but lane service time is continuous across it, with an estimated discontinuity of 1.7 s (95% CI –4.9 to 8.3) against a mean transaction of 96.4 s. Density testing indicates substantial manipulation of the running variable at the limit, with excess mass at exactly twelve; a donut specification excluding the three adjacent counts returns a comparable estimate. Nine stores reduced their posted limit from fifteen to twelve during the observation window, and we observe that the point of manipulation migrated within three weeks while service time did not move. Over the same period the chain’s express-lane integrity measure improved. Our results suggest that a posted limit may be better understood as a sorting instrument than as a service-time instrument, though the two are not always distinguished in front-end practice.

Introduction

An express checkout lane carries one number. It is printed on a hanging sign, it is not enforced by any mechanism other than the shopper’s own reading of it, and it has been in continuous use in grocery retail for longer than most of the front-end technology around it. The number is understood to be about speed. Whether it is about speed is a question the literature has largely declined to ask, in part because the setting resists experimentation: no chain will randomise the sign above a lane its customers are already queueing under.

The threshold form, however, supplies its own identification. A shopper arriving with eleven items is eligible for a lane that a shopper with thirteen is not, and — absent manipulation — the two are otherwise comparable. This is the

canonical setting for a regression-discontinuity design [1], [2], and it has the practical advantage that the assignment variable is recorded by the register regardless of which lane the basket ends up in.

We assemble lane-level transaction records from 22 stores of a single national grocery chain over nine consecutive weeks and estimate the discontinuity at the twelve-item limit in three outcomes: express-lane use, lane service time, and basket composition. Our contributions are as follows.

- We provide what may be the first regression-discontinuity estimate of an express-lane item limit’s effect on lane service time, although our identification rests on a threshold the chain set for reasons it did not record.

- We document substantial manipulation of the running variable at the limit and show that the resulting estimates appear robust to a donut exclusion, which we retain for completeness.
- We observe that a limit reduction implemented mid-study moved the point of manipulation without moving service time, a result that may generalise to other posted thresholds.

The finding we consider most consequential is the least eventful: the rule sorts, and sorting is not the same as speed.

Background

Queueing at a service point has been studied both as an operational quantity and as an experience. The psychology of waiting distinguishes the objective delay from its evaluation [3], and queue structure — how the line is arranged and what the waiting party can see of it — has measurable effects on attitudes independent of the wait itself [4], [5]. Queue state also shapes behaviour on the retail floor: longer visible queues are associated with changes in what shoppers buy [6], and store traffic bears on conversion in ways that complicate simple throughput accounts of front-end performance [7].

Service time itself is not a fixed property of a task. Server workload affects how long a task takes [8], and the load effect has been characterised across a range of service settings [9]. This bears directly on the interpretation of any lane-level comparison: a lane’s service time reflects the conditions under which it is worked, not only the basket in front of it.

Methodologically we follow standard practice for sharp designs [10], [11], using local linear estimation with robust bias-corrected inference [12] and avoiding high-order global polynomials [13]. Because our running variable is an integer chosen by the shopper, manipulation is a live concern; we test for it with density-based procedures [14], [15] and report a donut specification of the kind used where heaping at the threshold is suspected [16]. The behavioural pattern we observe — excess mass immediately below a salient threshold — is the same phenomenon documented at tax kinks and notches [17], [18] and reviewed as a general estimation strategy in [19].

Finally, the indicator side of our findings sits in a long-standing literature on the reactivity of published measures [20], [21], on target-driven

gaming in delivery organisations [22], and on the way public measurement reshapes the practice being measured [23]. Prior work at this institution has found that queue state can shift a classification decision made at a service gate [24], and that shoppers’ stated trust in a self-check-out error prompt does not predict their observed override behaviour [25].

Method

Setting and data

The chain operates express lanes signed with a posted item limit at all 22 study stores, distributed across six metropolitan regions. Signage wording varied across stores (“12 items or less”, “12 items or fewer”, and, at two stores, “About 12 items”); we treat all three as the same posted limit and report a wording-stratified estimate in the robustness set. Nine stores reduced their posted limit from fifteen to twelve in week four of the observation window as part of a chain-level front-end standardisation, which we exploit in Section 5.

The unit of analysis is the completed check-out transaction. We observe the register identifier, the lane class (express or standard), the scanned item count, the tender type, the count of weighed produce events, the count of price-check interventions, and two timestamps: first scan and payment complete. Lane service time is the interval between them. Four exclusions are applied in order: baskets outside the estimation window, transactions held open by a supervisor, self-service lanes, to which no posted limit applies, and staff purchases. After these the analysis set contains 38,412 transactions.

```
# exclusions, applied in order
drop if item_count > 40
drop if service_time_s > 600
drop if lane_class == "self"
drop if tender == "staff_account"
```

Estimation

For running variable R_i (scanned item count), cutoff $c = 12$, and outcome Y_i , the sharp estimand is

$$\tau = \lim_{r \downarrow c^+} \mathbb{E}[Y_i \mid R_i = r] - \lim_{r \uparrow c^-} \mathbb{E}[Y_i \mid R_i = r]$$

estimated by local linear regression fitted separately either side of the cutoff with a triangular kernel and a data-driven bandwidth, with robust bias-corrected confidence intervals [12].

Because R_i is discrete and the support either side of the cutoff is modest, we cluster standard errors by store-week and report estimates under six specifications. All estimates were pre-specified in a protocol lodged with the School before the transaction extract was received; the protocol fixed the cutoff, the bandwidth selector, and the exclusion list, and did not anticipate the mid-study limit reduction.

Results

The limit sorts

The first stage behaves as the sign intends. The probability that a transaction is processed in an express lane falls by 54.8 percentage points at the cutoff (95% CI 50.1 to 59.5). Compliance with the posted limit is high and asymmetric: baskets above the limit almost never appear in an express lane, while baskets well below it frequently choose a standard one.

Service time does not move

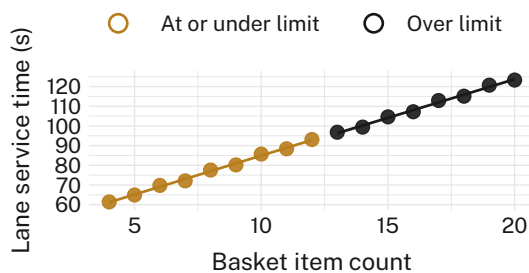


Figure 1: Lane service time rises by roughly 3.9 s per additional item and does not step at the twelve-item limit; the fitted values either side of the cutoff differ by 1.7 s (95% CI -4.9 to 8.3).

Lane service time is continuous across the cutoff. The estimated discontinuity is 1.7 s (95% CI -4.9 to 8.3) against a mean transaction of 96.4 s, a difference smaller than the standard deviation of a single price-check intervention. Item count is a real but partial predictor of service time: within store, it accounts for 18.4% of the variance, while tender type, weighed produce events, and price checks together account for 46.9%. The attribute the sign can carry is not the attribute that drives the outcome the sign is understood to manage.

Shoppers manipulate the running variable

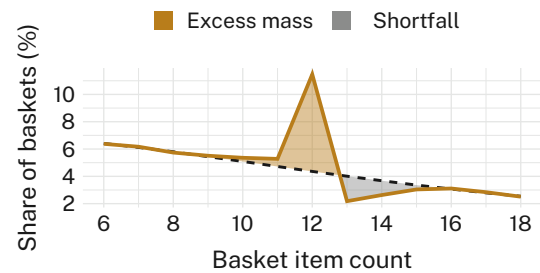


Figure 2: Observed basket counts show excess mass at exactly twelve against the density fitted from outside the manipulation window, with a compensating shortfall at thirteen and fourteen; the bunching ratio is 2.63 (95% CI 2.11 to 3.15).

Density testing rejects continuity of the running variable at the limit ($p < 0.001$), with excess mass at exactly twelve and a compensating shortfall immediately above it. The bunching ratio is 2.63 (95% CI 2.11 to 3.15). We interpret this as basket adjustment at the point of lane choice, an interpretation consistent with the shortfall's location. A donut specification excluding counts eleven through thirteen returns an estimated discontinuity of 2.2 s (95% CI -5.6 to 10.0), which we take as reassuring.

Specification and ablation

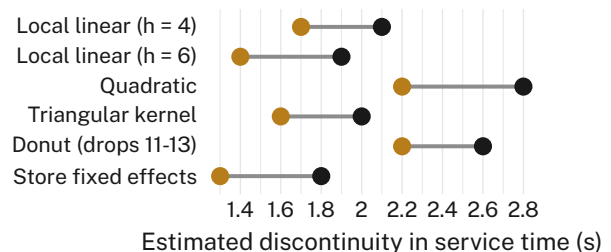


Figure 3: Removing the basket-composition controls shifts every estimate by less than 0.6 s and no specification's interval excludes zero; the controls are retained for completeness.

Table 1 summarises the primary estimates. Removing the basket-composition controls degrades nothing: the estimated discontinuity moves by at most 0.6 s across the six specifications and no interval excludes zero. We retain the controls, and note that their retention is not consequential.

Outcome	Est.	95% CI	n
Express-lane use (pp)	-54.8	-59.5, -50.1	38,412

Outcome	Est.	95% CI	n
Service time (s)	1.7	−4.9, 8.3	38,412
Service time, donut (s)	2.2	−5.6, 10.0	31,884
Basket items	−0.1	−0.4, 0.2	38,412
Card tender share (pp)	0.6	−1.9, 3.1	38,412

Table 1. Primary estimates at the twelve-item cutoff. Standard errors clustered by store-week.

The threshold moves; the outcome does not

The nine stores that reduced their posted limit from fifteen to twelve provide an unplanned test. The modal point of excess mass migrated from fifteen to twelve within three weeks of the sign changing, and the bunching ratio at the new limit was indistinguishable from the ratio at the old one. Mean express-lane service time at those stores changed by 0.8 s (95% CI −3.1 to 4.7) over the same period.

Across the nine stores the chain's express-lane integrity measure — the share of express-lane transactions at or under the posted limit — rose from 88.1% to 96.4%. In the chain's front-end reporting the limit reduction is recorded as a completed queue-time action, and the improvement in the integrity measure is attributed to it.

Discussion

We observe consistent results across specifications: the posted limit is an effective sorting instrument and is not, on our estimates, a service-time instrument. The two functions are easily conflated because they share a sign. A limit sorts baskets by the only attribute a hanging sign can carry, and item count is the attribute least predictive of the outcome the lane exists to manage.

The mid-study limit reduction is instructive precisely because it changed so little. Lowering the number moved the point at which shoppers put an item back, raised the share of transactions in compliance with the posted limit, and left service time where it was. An indicator defined on compliance will improve under such a change by construction, and will continue to improve as the number falls. This is the reactivity the measurement literature describes [20], [23], arriving through an unusually short causal path: the measure counts adherence to a sign, and the sign is the one thing management can adjust without touching staffing.

We do not read our null as evidence that express lanes are without value. Sorting has its own

benefits, including a shorter and more legible queue for the shopper who qualifies, which may bear on evaluations of the wait independent of its length [3]. Our estimates speak only to the service time of the transaction itself.

Limitations

Our evaluation is limited to one chain, though the consistency across 22 stores and six metropolitan regions suggests broader applicability. We observe only transactions that reached a register, so shoppers who left without purchasing are outside the design; the rate of such departures is low in this format. Manipulation of the running variable is substantial, which is itself one of our findings, and the donut specification returns a comparable estimate. Finally, we cannot observe why the chain set its limit at twelve, a gap that is arguably the study's most useful result.

Conclusion

An express-lane item limit sorts baskets sharply and moves lane service time by an amount our design cannot distinguish from zero. Shoppers manipulate their baskets to sit at the limit, and when the limit falls they manipulate to the new one, leaving the outcome unchanged and the compliance measure improved. Future work might examine whether a limit posted in expected seconds rather than in items would sort on the attribute that matters, and whether any shopper could read it.

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